

Final Project Report

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Introduction

When people talk about trash pollution, the conversation usually stays abstract—“plastic in the ocean,” “litter on the beach,” “microplastics everywhere.” This dataset makes the problem concrete. It records individual dumpster collections over time (2014–2023), tracking weight (tons), volume (cubic yards), and item counts like plastic bottles, cigarette butts, wrappers, and plastic bags.

What stood out immediately is that the same story can look different depending on *which metric you graph*. Weight and volume don’t always move together, and “how much trash” can mean mass, space, number of trips, or even the “Homes Powered*” estimate tied to trash-to-energy framing.

Exploratory Data Analysis (EDA)

What the dataset contains

Each row represents one dumpster collection event with:

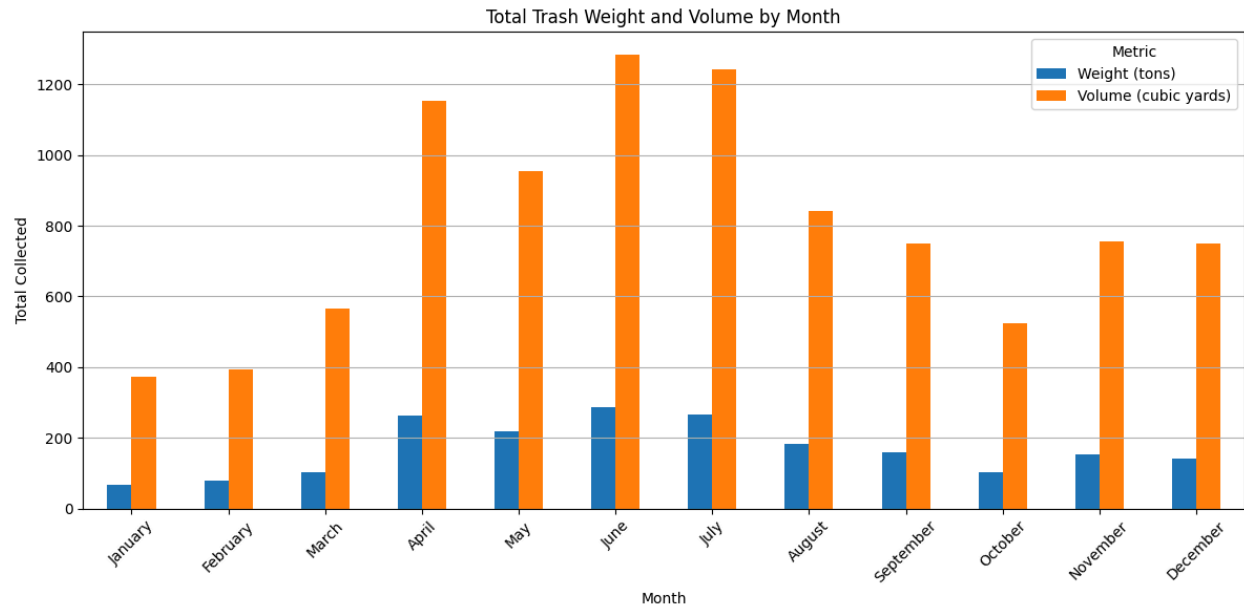
- Time variables: Month, Year, Date
- Core totals: Weight (tons), Volume (cubic yards)
- Item counts: Plastic Bottles, Polystyrene, Cigarette Butts, Glass Bottles, Plastic Bags, Wrappers, Sports Balls
- Derived metric: Homes Powered*

The totals row suggests the full time span includes 2,019.54 tons of trash across 9,589 cubic yards, and 11,735,100 cigarette butts (by far the largest item count).

Seasonality and Monthly Patterns

Weight vs. volume by month

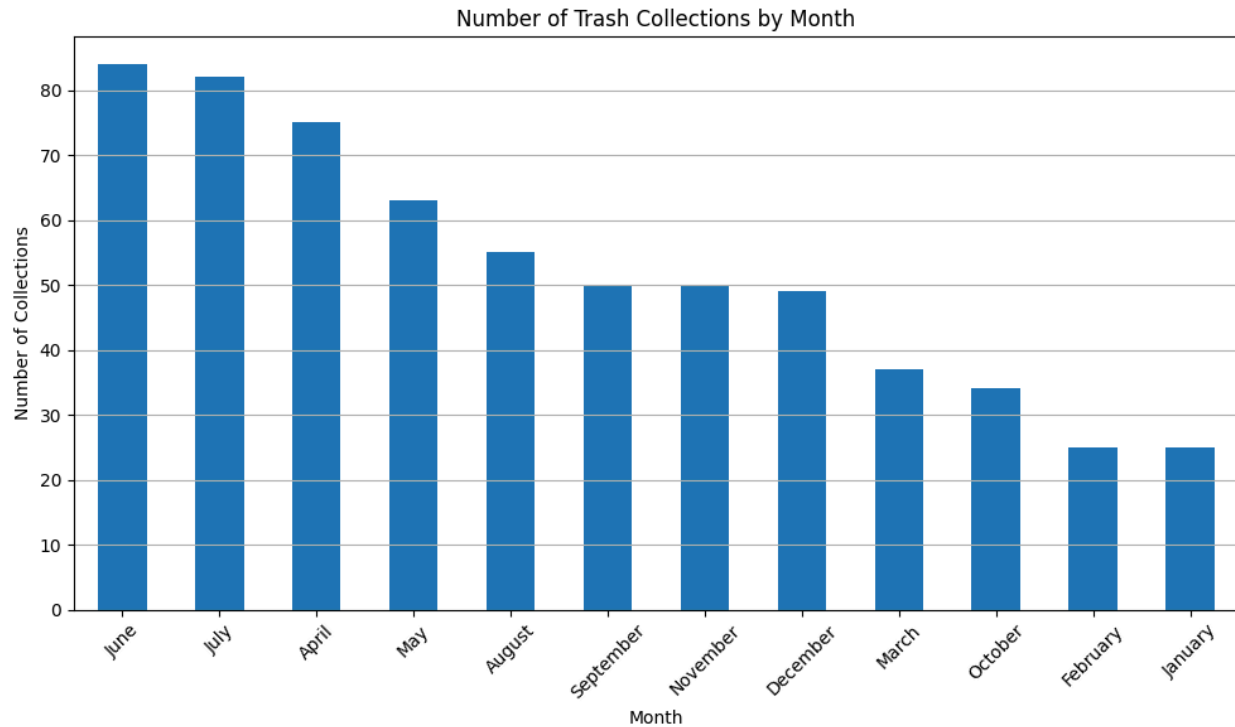
The monthly totals chart shows a clear “trash season.” April–July dominate both weight and volume, with the biggest volumes clustered in the summer months.



A key point: volume spikes can reflect bulky/light materials, while weight spikes reflect denser loads. Looking at only one of these can hide what’s actually happening in the dumpster.

Collection workload by month

The collections-by-month chart adds another layer: the busiest months are June and July, which helps explain why total trash also peaks then—more trips usually means more total material collected.

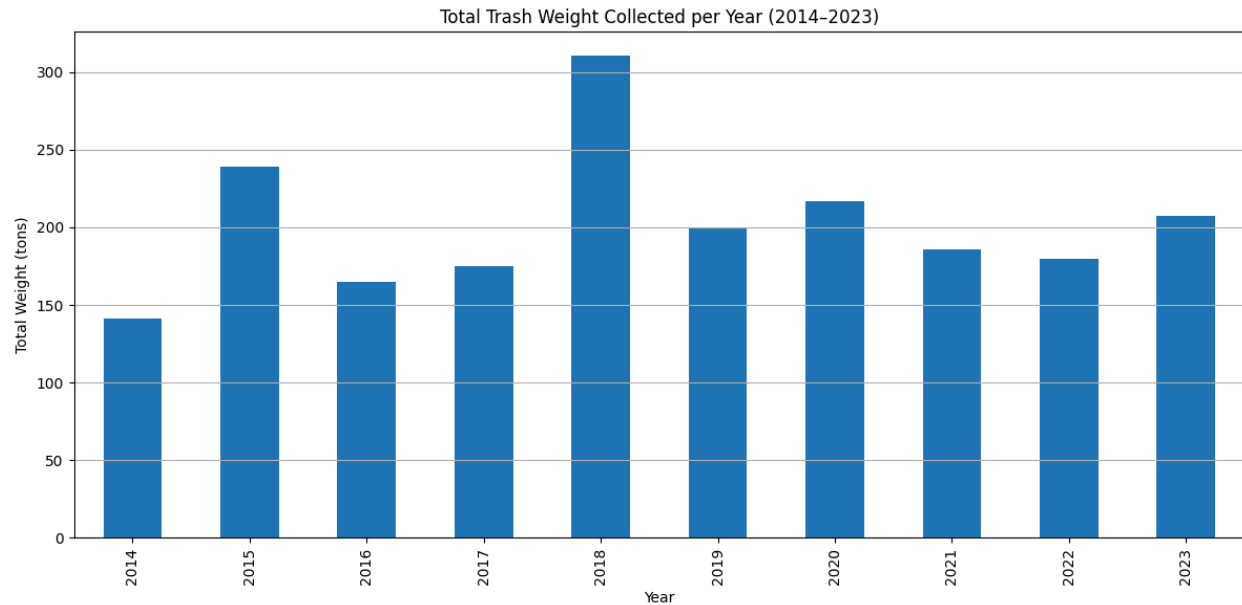


If your question is operational (staffing, scheduling, volunteer coordination), this chart is often more useful than weight alone.

Year-to-Year Trends (2014–2023)

A standout peak year

Looking at total trash weight by year, there's a dramatic peak around 2018, followed by a drop and then more stable totals with smaller rebounds later.

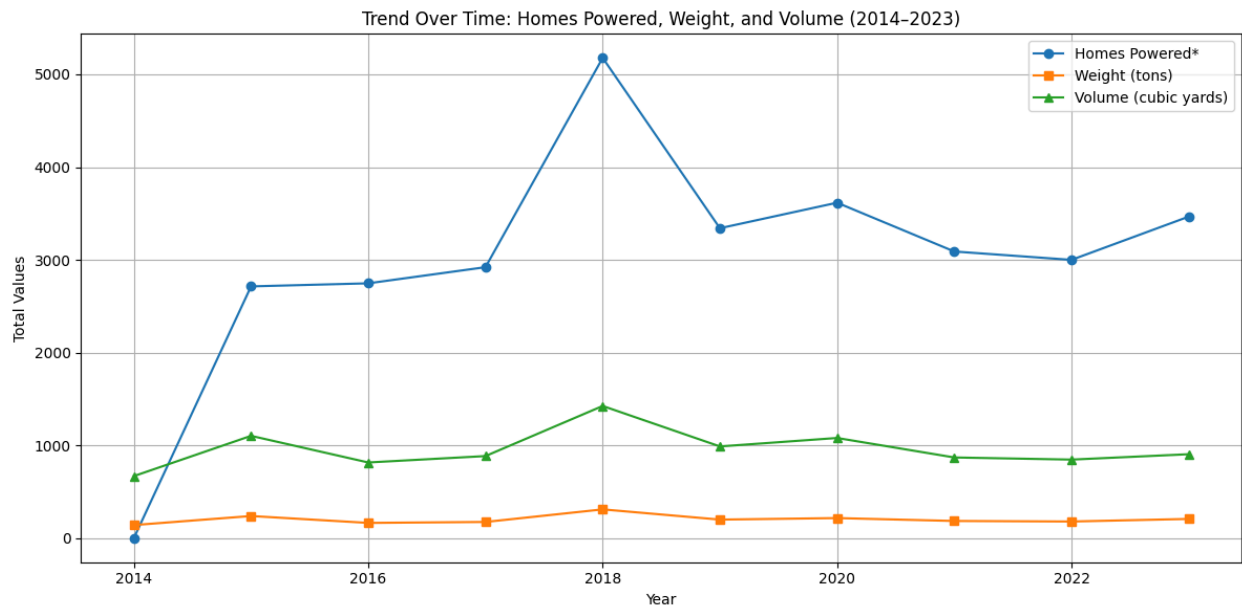


This spike could reflect a real increase in trash, an expansion in collection effort, or changes in where/how collections were done. The chart tells us *what* happened—not *why*—but it clearly marks 2018 as an outlier worth investigating.

Homes Powered* vs Trash Metrics

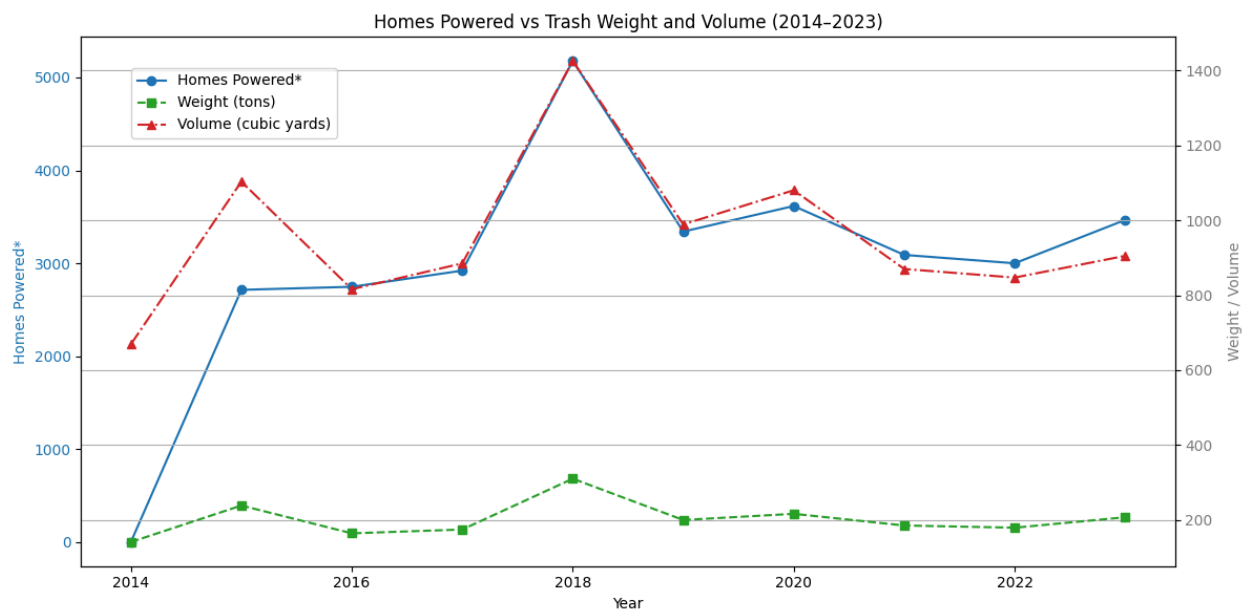
How Homes Powered* tracks totals over time

When plotted alongside weight and volume, Homes Powered* rises and falls in a similar shape—especially around the peak year—suggesting it scales with total trash collected.



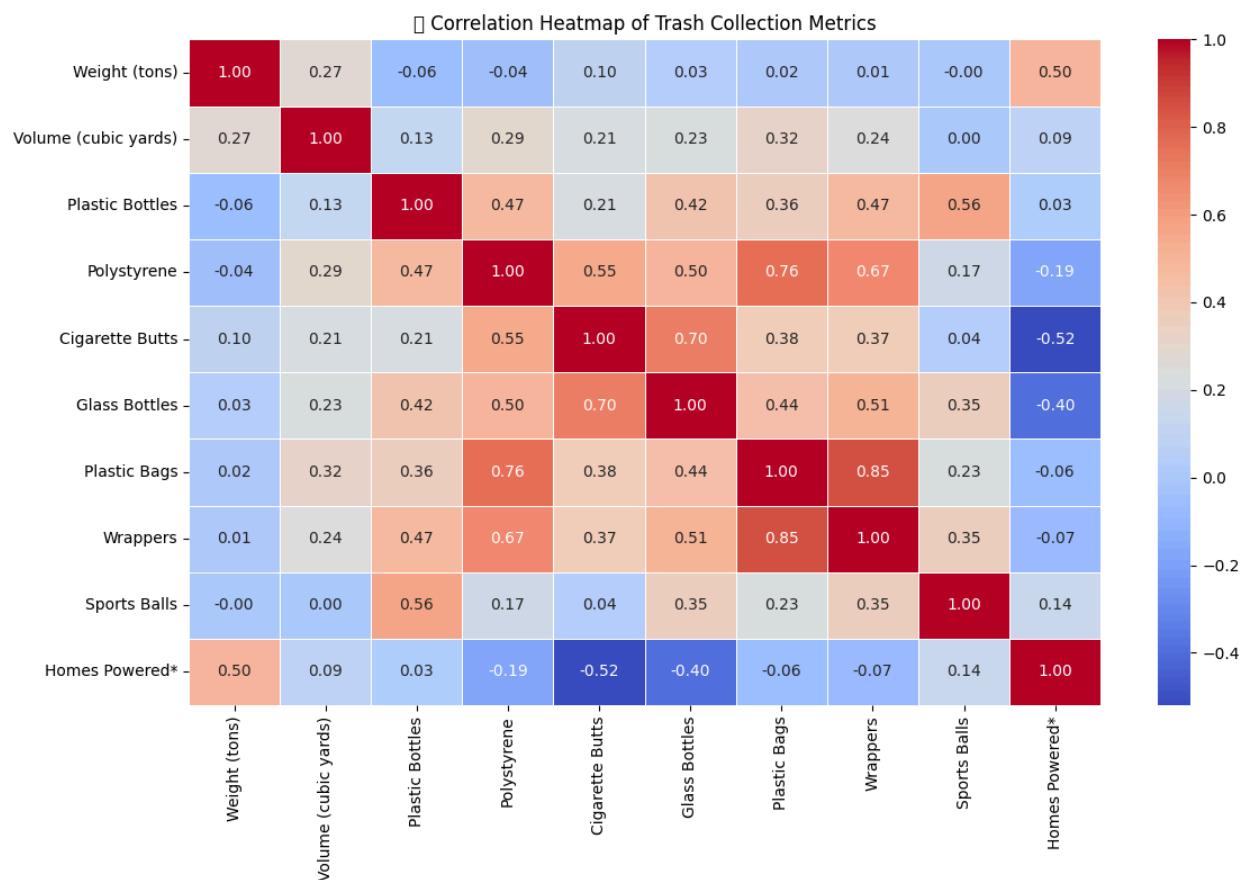
Same relationship, different view

This combined plot reinforces the idea that Homes Powered* follows trash totals, but it also emphasizes that weight and volume don’t always behave identically—and Homes Powered* tends to “feel” more aligned with weight than volume.



Correlation Matrix Findings

The correlation heatmap helps answer: *which variables truly move together?* It also reveals when “obvious” relationships aren’t actually that strong.



Key takeaways from the heatmap:

- Plastic Bags ↔ Wrappers (~0.85) is very strong, suggesting a shared “packaging litter” stream.
- *Homes Powered* ↔ *Weight* (~0.50)* is moderately positive, while *Homes Powered* ↔ *Volume* (~0.09)* is weak—consistent with the idea that energy equivalence would depend more on mass than space.
- Homes Powered* has negative relationships with Cigarette Butts and Glass Bottles, which likely reflects *differences across time periods or reporting*, not that those items “reduce” energy. This is a good reminder that correlation can capture program changes, measurement shifts, or time effects.

Conclusion

Key Findings

- Trash collection is strongly seasonal, peaking in late spring and summer.
- 2018 is the highest-weight year, and it stands out enough to deserve a deeper explanation.
- Homes Powered* tracks overall trash totals and aligns more with weight than volume.
- Some trash types appear linked (especially plastic bags and wrappers), suggesting consistent composition patterns in many collections.

Further Investigation

Next steps that would add a lot of insight:

1. Compute average weight/volume per collection (to separate “more trash” from “more trips”).
2. Compute item counts per ton (composition changes over time).
3. Split analysis into early vs later years to check whether Homes Powered* is calculated consistently.
4. Cluster collections into types (“packaging-heavy,” “butt-heavy,” etc.) to see what patterns drive the peaks.